

PATRICK LESSA

Data Scientist & Full Stack Developer

Python · SQL · LLMs & RAG · Machine Learning · Apache Airflow · Flask / React · Power BI

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PROFESSIONAL SUMMARY

Data and applied-AI professional with 8+ years turning multi-source data into business decisions, production pipelines, executive dashboards, and GenAI products. I work end to end, from problem modeling through delivery and documentation, with Python, SQL, Apache Airflow, Docker, Spark/PySpark, Microsoft Fabric, and Power BI, plus hands-on experience with LLMs, RAG, embeddings, and AI observability. I build from the backend (Python, Flask, REST APIs) to the frontend (React), and translate ambiguous business demands into KPIs, architecture, and maintainable products. Available for remote work.

CORE SKILLS

Languages	Python (Pandas, NumPy, scikit-learn, TensorFlow), SQL, JavaScript, HTML, CSS
AI / LLM	RAG, LangChain, embeddings, semantic and vector search, pgvector, hybrid retrieval, reranking, prompt engineering, grounded generation, document intelligence, multi-provider orchestration (OpenAI, Gemini, Claude)
ML & Statistics	Segmentation, clustering, classification, regression, ensemble methods, regularization, cross-validation, probabilistic models (GMM, Markov chains), time series, factor analysis / PCA, CFA/SEM
Data Engineering	ETL/ELT, production pipelines, workflow automation, multi-source integration, validation, migration
Orchestration	Apache Airflow, Docker, Spark / PySpark, Git
Platforms & Cloud	Microsoft Fabric (Lakehouse, Dataflows Gen2, Spark notebooks), Azure SQL, Azure, PostgreSQL / Supabase; GCP / BigQuery (transferable); AWS (expanding)
Backend & Frontend	Flask, REST APIs, React
BI & Visualization	Power BI (DAX, Power Query, star schema, RLS), semantic modeling, KPI layers, Plotly
Power Platform	Power Apps, Power Automate, SharePoint, Dataverse
AI Governance	Traceability, hallucination control, fallback strategies, observability of latency, cost, and answer quality

PROFESSIONAL EXPERIENCE

Senior Data & Analytics Scientist Jul 2024 – Present

Mouts TI · Remote (official title: Senior Power Platform Developer · clients: Atlas Renewable Energy, ENGIE, Starrett, credi&gente)

- Architected and maintain end-to-end production data pipelines with Apache Airflow and Docker, orchestrating daily ingestion, transformation, and delivery of multi-country market and valuation data, with anomaly checks in the first 30 minutes of every run.
- Automated ingestion of **14 SharePoint and SQL sources**, cutting monthly data preparation from 6 hours to 45 minutes, an **85% efficiency gain**.
- Delivered executive risk dashboards for LATAM operations **in under 30 days**, later scaled to a global rollout; they informed board-level decisions on curtailment, spot pricing, and trading.
- Built Python integrations against REST APIs, SharePoint, and the Zeus ETRM platform, replacing manual routines with reliable, observable, and documented flows.
- Modeled scalable analytical layers in Microsoft Fabric (Dataflows Gen2, Lakehouse, Spark notebooks) with star schema and per-country row-level security; authored the technical documentation.
- Migrated the Starrett backend from SharePoint Lists to Azure SQL and adapted the Power Apps layer, improving reliability, scalability, performance, and user experience.

Data Analyst and Project Manager Apr 2023 – Jul 2024

TD Business · Remote

- Led and co-led the development of intelligence centers for public agencies and strategic institutions, including the Ministry of Education, the Brazilian Space Agency, FEESC, and Sebrae.
- Contributed to delivering **over 100 BI dashboards in roughly 15 months**, defining indicators, datasets, and reporting logic that became the source of truth for stakeholders.
- Helped structure the Sebrae Startups Observatory, running cohort analyses to map innovation ecosystems and produce executive reports.
- Translated loosely structured business demands into KPIs, data models, and decision-oriented dashboards.

- Mapped and instrumented **363 institutional processes in 2 months**, exposing operational gaps and creating an analytical baseline.
- Developed **20+ strategic KPIs** for the Distance Education division and automated 10 manual spreadsheets, compressing update cycles from weeks to days.

- Ran market research and behavioral analysis, and built segmented campaigns from performance data and customer feedback.
- Supported decision-making with data-driven recommendations, the foundation of the storytelling later applied in analytics.

INDEPENDENT PRODUCT

- Design, build, and maintain the entire platform single-handedly: **19 tools in production**, roughly **89,000 lines of Python**, and **947 automated tests**.
- Built a RAG research assistant in Python: multi-provider orchestration (OpenAI, Anthropic, Google) normalized into a single streaming protocol with fallback chains, hybrid retrieval combining keyword and semantic similarity, reranking, and answers grounded in traceable citations anchored to the exact passage of the source PDF.
- Engineered vector search on PostgreSQL with pgvector and an HNSW cosine index, plus diversified retrieval resolved in SQL so no single document dominates the model context.
- Implemented multi-step AI workflows: query classification and rewriting, chunking, caching, fallback chains, context persistence, streaming, and uncertainty handling.
- Shipped the surrounding SaaS: Stripe subscriptions with idempotent webhooks, atomic credit debits inside Postgres, row-level security, structured logging with PII masking, Docker, and CI on GitHub Actions.

QUANTITATIVE MODELING & APPLIED RESEARCH

- Python-based analysis, market segmentation, and probabilistic modeling with Gaussian Mixture Models and Markov chains.
- Statistics, exploratory factor analysis, PCA, time series, CFA/SEM, and experimental design, applied across doctoral research and consumer behavior studies.

EDUCATION

PhD in Business Administration · in progress	2024 – Present
Universidade Federal de Santa Catarina (UFSC) · Data-driven marketing, predictive segmentation, probabilistic models	
Master's in Business Administration · CAPES scholarship	2022 – 2024
Universidade Federal de Santa Catarina (UFSC) · Mixed methods, web scraping in Python, exploratory factor analysis	
Bachelor's in Business Administration · PIBIC/CNPq research scholarship	2018 – 2021
Universidade Federal do Cariri (UFCA) · Consumer behavior, Power BI, quantitative methods	

CERTIFICATIONS & LANGUAGES

Certifications	Google Data Analytics · Python Data Analytics · Supervised Machine Learning · Microsoft Power BI · Advanced Excel
Languages	Portuguese (native) · English (advanced, B2–C1) · Spanish (basic) · French (basic)